Food Packaging- Lesson 2

Printing of packages, barcodes and other markings

Printing

- It is a process in which text and images are reproduced, typically with ink on substrate using a printing press made from letters, photographs and drawing.
- The major printing processes are distinguished by (i) the method of image transfer and (ii) by the general type of image carrier employed.

I. Depending upon the process (the printed image is transferred to the substrate either directly or indirectly):

- In **direct** printing the image is transferred directly from the image carrier to the substrate. Examples of direct printing are gravure, flexography, screen printing and letterpress printing processes.
- In **indirect**, or offset, printing, the image is first transferred from the image carrier to the blanket cylinder and then to the substrate. Lithography, currently the dominant printing technology, is an indirect (offset) process.

II. Depending on the image carriers employed:

- Image carriers (or plates) are generally be classified as relief, planographic, intaglio, or screen.

➢ In **relief printing**, the image or printing area is raised above the non-image areas. Letterpress and flexography are relief printing processes.

➢ In **planographic printing**, the image and non-image areas are on the same plane. The image and non-image areas are defined by differing physiochemical properties. *Lithography* is a planographic process.

➢ In the **intaglio process**, the nonprinting area is at a common surface level with the substrate while the printing area, consisting of minute etched or engraved wells of differing depth and/or size, is recessed. Gravure is an intaglio process.

➢ In the **screen process** (also known as porous printing), the image is transferred to the substrate by pushing ink through a porous mesh which carries the pictorial or typographic image. Screen printing is a porous printing.

•Offset Printing:

- A widely used printing technique where the inked image is transferred (offset) from a plate to a rubber blanket, then to the printing surface.
- Books, newspapers, brochures, stationery.

•Digital Printing:

- A method of printing from a digital-based image directly to a variety of media. It usually refers to professional printing where small-run jobs from desktop publishing and other digital sources are printed using large-format and/or high-volume laser or inkjet printers.

Short runs of books, flyers, business cards, personalized printing.

•Screen Printing:

- A printing technique where a mesh is used to transfer ink onto a substrate, except in areas made impermeable to the ink by a blocking stencil.
- T-shirts, posters, signage, ceramics, glass, metal, wood

•**Flexography**:

A form of printing process which utilizes a flexible relief plate. It is essentially a modern version of letterpress and can be used for printing on almost any type of substrate, including plastic, metallic films, cellophane, and paper.

Food packaging, labels, wall coverings, gift wrap.

Gravure Printing:

: An intaglio printing process, where the image is engraved onto a cylinder. The ink is filled into the engraved areas of the cylinder and transferred onto the paper.

Magazines, postcards, packaging, wallpapers.

Letterpress Printing:

The oldest form of printing text, in which a raised surface (like metal type or a plate) is inked and then pressed to a substrate, usually paper.

•Stationery, wedding invitations, business cards, fine art prints

3D Printing:

- The process of making a three-dimensional solid object from a digital file by laying down successive layers of material.
- Prototyping, manufacturing, medical devices, art and design

Lithography:

- A printing process based on the repulsion of oil and water. The image area is treated to retain ink, while the non-image areas repel ink.
- Posters, maps, packaging, books.

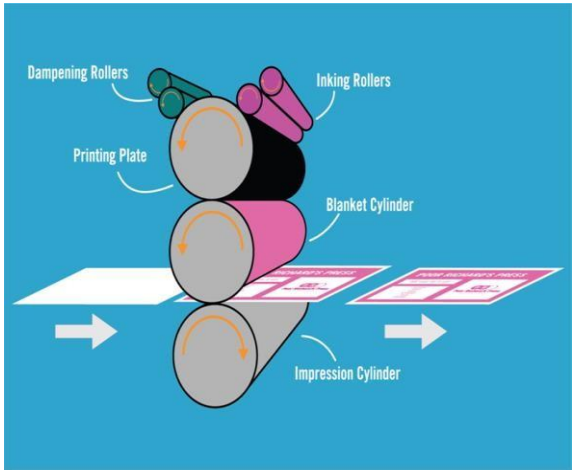
Thermal Printing

- A digital printing process which produces a printed image by selectively heating coated thermochromic paper (thermal paper) when the paper passes over the thermal print head.
- Receipts, shipping labels, barcodes.

Pad Printing:

- An indirect offset (gravure) printing process, where a silicone pad picks up ink from an etched plate and transfers it to a substrate.
- Printing on irregular or curved surfaces like pens, bottles, and electronic

1. Lithography

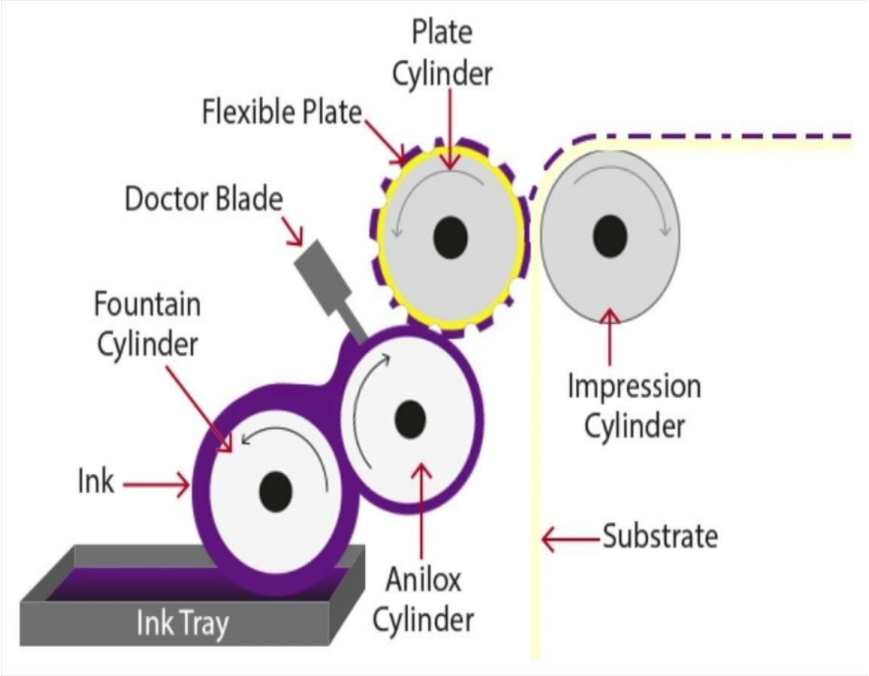


- Lithography is a method in which printing is applied on a smooth surface using chemical processes.
- In this method, the positive part of an image would be a hydrophobic chemical, while the negative image would be water.
- Thus, the image areas of the surface will be receptive to ink, and the non-image areas will repel ink and be receptive to water.
- The image is first set off from the plate to the blanket and then set off from the blanket to the paper. For this reason, lithography is also called **offset printing**.
- Commonly used for large or flat items such as folding cartons, corrugated packaging, and large labels used on packaging and retail displays.

- The lithographic process operates with three basic cylinders. They are the plate cylinder, the blanket cylinder, and the impression cylinder. All these are plain heavy metal cylinders.

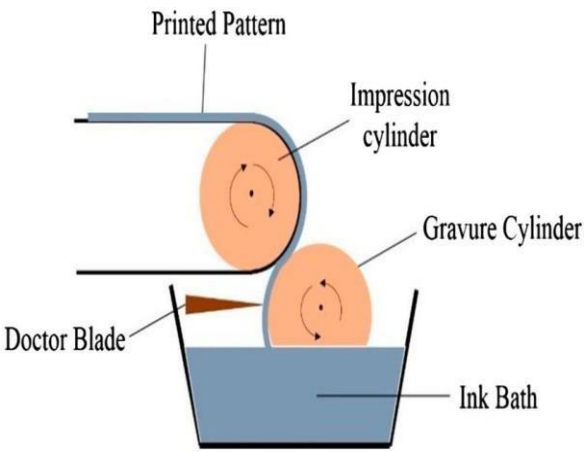
1. The *plate cylinder* has the printing plate wound around it. This plate is the carrier of the image that needs to be printed.
2. The *blanket cylinder* has a rubber blanket wound around it. This facilitates the transfer of the image from the plate to the blanket, and thereupon to the paper (or other substrates), when the substrate is passed between the blanket and the impression cylinder.
3. The *impression cylinder* is just a bare cylinder that acts to provide the necessary pressure to impress the image from blanket to the substrate.

2. Flexography



- It gets its name from the use of a flexible relief plate in the process and is often used in the printing of food packaging.
- Flexography is a direct process where the rubber printing plate transfers ink directly to the printing surface.
- This process is similar to relief printing. The image is first engraved onto the plate with a laser.
- Then the ink is transferred from its chamber to an anilox roller where a blade removes the excess ink.
- After this is completed, the ink is pressed onto the printing surface, and the product is coated in the desired finishing

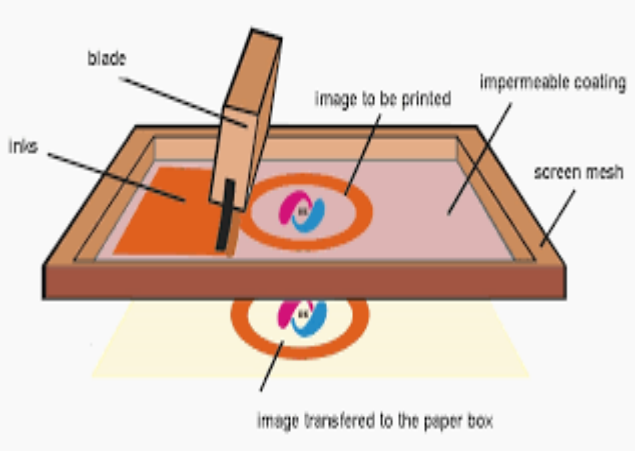
3. Rotogravure



- Gravure is a direct and rotary technique of printing on packaging.
- Prints from wrong-reading recessed image cylinder to substrate directly. Hence, it is also known as <u>recess printing</u>.
- This technique is ideal for very high volumes.
- This process includes a rolling cylinder which is engraved with the image that has to be printed.
- Once completed, it is inked and pressed directly onto the print surface to transfer the image.
- The rolling cylinder has different cell depths for ink.
- This means that deeper cells will have more ink and will be more intense in that print area, whereas a shallow cell with produce a lighter outcome.

4. Screen printing

- It is a pretty simple process to understand and operate
- The process derives its name from this silk screen, which was used as the image carrier.
- The image that needs to be printed is first captured on a photographic material, a positive usually.

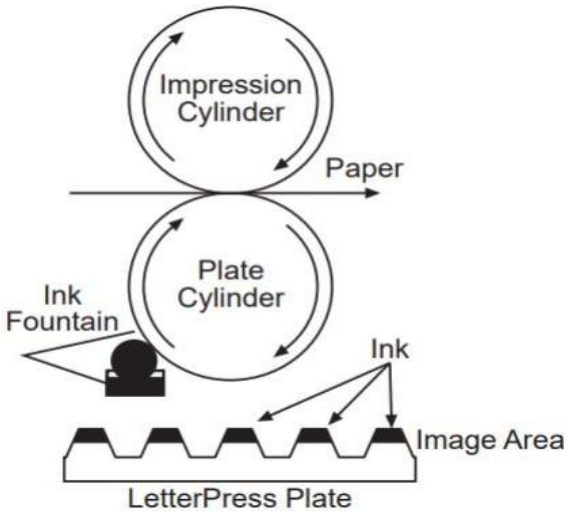


- The positive image is then transferred to the screen, which is on the porous area of the screen.
- The non-image areas are blocked out during the stage of image creation itself.
- The screen is laid over the substrate that is to be printed and ink is poured on the frame over the screen.
- The ink is then wiped across the surface of the screen using a device called a squeeze, a wooden device that has a rubber blade.

- Since the screen is porous in nature, the ink flows through it. Because the image areas are porous, they allow ink to flow through them. This ink is thus printed onto the substrate beneath

- Principal applications: can print on any substrate; point-of purchase displays, billboards, decals, fabric, electronic circuit boards, glasses, etc.

5. Letter press/Relief printing



- It is the earliest form of printing and remained dominant for a very long time.

- As the name of the process says, the image areas are in relief and the non-image areas are in recess.

- On application of ink, the relief areas are coated with a film of ink and the non-image areas are not.

- With pressure over the substrate to bring it in contact with the image area, the image is then transferred to the substrate.

- It is widely used for printing on the non-porous substrates required for various types of food packaging (it is also well suited for printing large areas of solid color).

Modern technology in printing: Digital printing

- A design is created on a computer; it may be an individual design or replicated to give several hundreds of impressions.
- The ink, usually in powder form, is attracted on to the film surface and cured in place.
- Special coatings are necessary to receive the ink.
- A standard heat-sealable coating on the reverse side allows the film to be made immediately into packages.
- The system as yet is only suitable for narrow web widths and is capable of producing test packages for market research or promotional campaigns.

Barcodes

- A bar code may be defined as a series of bars and spaces arranged according to the encodation rules of a particular specification in order to represent data.
- Its purpose is to represent information in a form that is machine-readable.
- Bar codes are read by scanning devices that are programmed to analyze the structure of the bars and spaces and transmit the encoded data in electronic format.

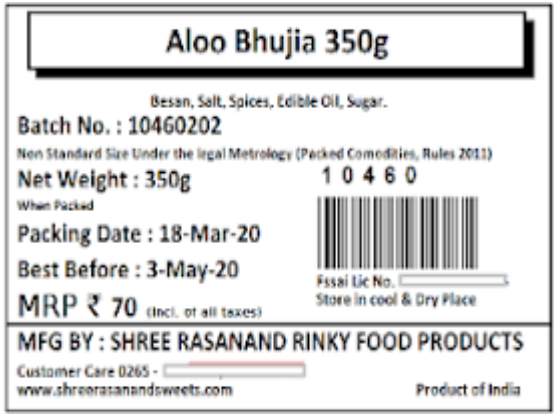
Barcodes on food packaging:

1. Help eliminate food waste in the supply chain

2.Improve the amount and quality of the information available about the product

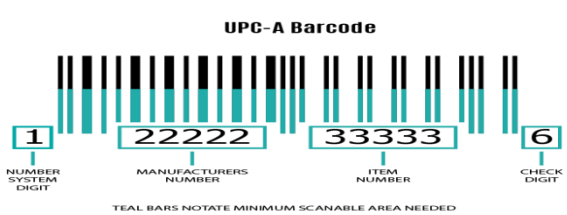
3.Maintain high levels of safety of consumable products

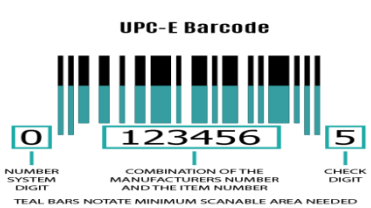
4. Provide accurate location and quality data about the product.



Types of Barcode

- There are several different Barcode Present that serve different uses, industries or geographic needs.
- Different barcode applications have different data requirements depending on their usage.
- There are one-dimensional and two-dimensional bar code types – UPC (Universal Product Code), EAN (European Article Number) are the two *one-dimensional codes* widely used in the retail industry.
- One-dimensional (or 1D bar codes) represent data by varying widths and spacing of parallel lines, and may be referred to as one-dimensional.
- These include most of the traditional widely used bar code types in the retail industry such as UPC and EAN bar codes.
- UPC bar codes are used to label and scan consumer goods at points-of-sale - mainly in the United States.





- EAN bar codes are used to label and scan consumer goods at points-of-sale — mainly in the Europe. They look very similar to UPC codes, and the main distinction is their geographical application.
- GS1 provides standards for use with different bar code symbols. The most widely used barcode is GTIN (Global Trade Item Number), a new naming structure for retail related barcodes.

- GTINs include:

- GTIN-12 (UPC-A): this is a 12-digit number used primarily in North America
- GTIN-8 (EAN/UCC-8): this is an 8-digit number used predominately outside of North America
- **GTIN-13** (EAN/UCC-13): this is a 13-digit number used predominately outside of North America
- GTIN-14 (EAN/UCC-14 or ITF-14): this is a 14-digit number used to identify trade items at various packaging levels

- GS1 India is a not-for-profit , industry led global standards body headquartered at Brussels, Belgium and was setup by Ministry of Commerce and Industry, Govt. of India

- UCC-Uniform Civil Code
- ITF- Interleaved Two of Five – mostly used to label secondary packaging

QR Code

- QR code, also known as 'Quick Response' code, is a *two-dimensional barcode* that can store information/URL/data.
- QR codes stock large data files and use four standardized encoding modes (numeric, alphanumeric, byte/binary, and kanji).
- Adding a QR code on the packaging is an effective communication tool
- Because of the QR code's fast readability and storage capacity, it is predominantly used by food industries in compliance with food regulatory measures.
- One can instantly access and save the PDF document by scanning the code.
- It adds a digital element to ordinary food packaging while ensuring that consumers have access to important nutritional information.

Advantages of QR Code on Food Packaging

- ✓ Enhancing Food Safety - QR codes play a crucial role in improving food safety by enabling traceability throughout the supply chain.
- ✓ Access to Nutritional Information - QR codes on food packaging provide a gateway to detailed nutritional information, including ingredients, allergens, calorie counts, and dietary guidelines.
- ✓ Boosts marketing - QR codes boosts product marketing by giving the possibility to add online social platforms, video links about the product, multilingual product documents, and costumer feedback and support .